PORTFOLIO

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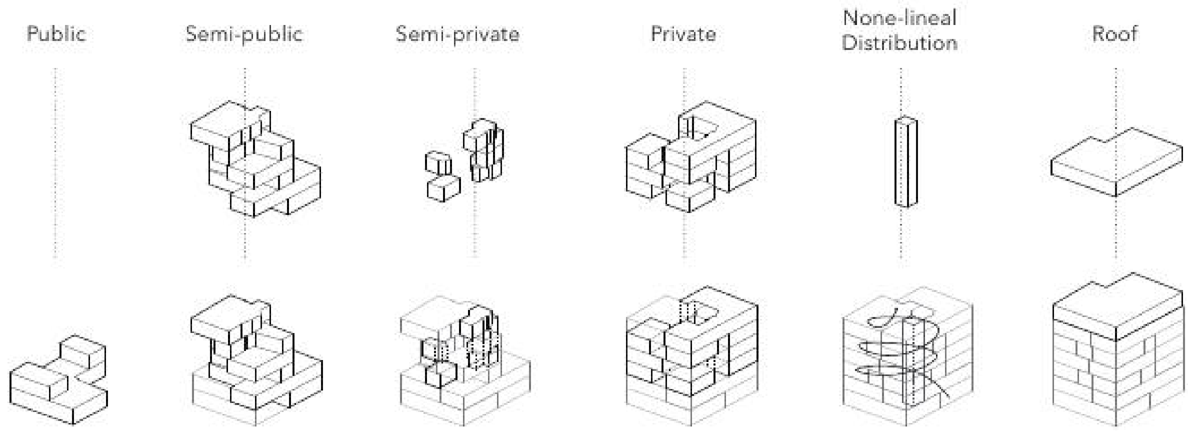
2014-2019
B.A. Architecture
Technical University of Munich

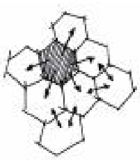
CONTENTS

01
HOW DO WE LIVE?
ROSENHEIMER STR. 66

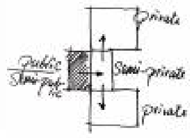
Shared Living in Haidhausen
2016 | Chair of Urban Design and Housing
Concept: Initially Group Work of Four, TU Munich
Redesign & Development: Single Work
Munich, Germany

We would like to discuss the issue of a new version of collective living in a single building in the center of Munich and to find a solution to it. This solution is relevant in the context of the sustainability debate, as the sharing of resources and minimizing of repetition represents resistance to the micro-flat culture.

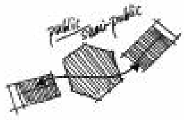




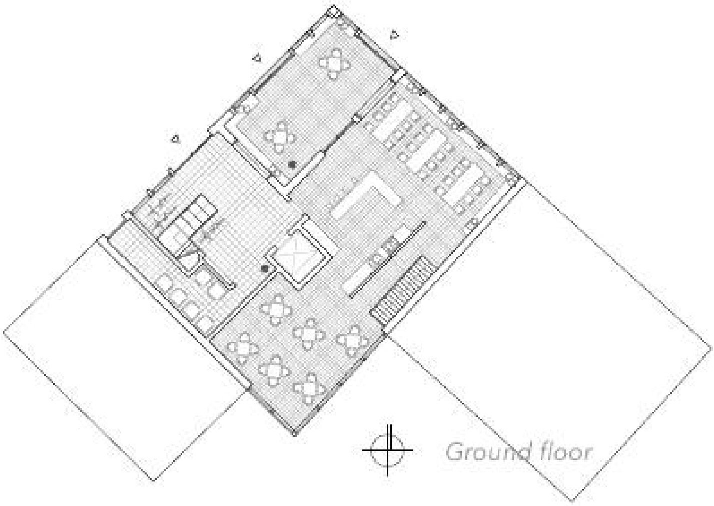
Continuous circulation through rooms instead of corridors



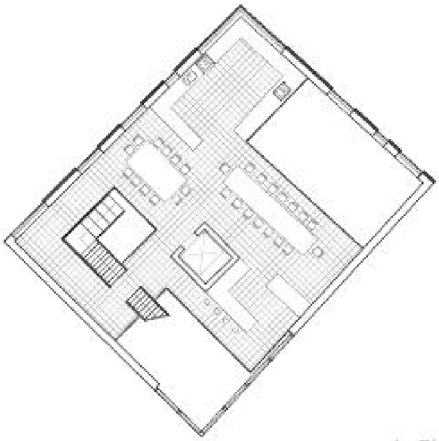
Hierarchical room division



None-lineal distribution: common room as distribution area



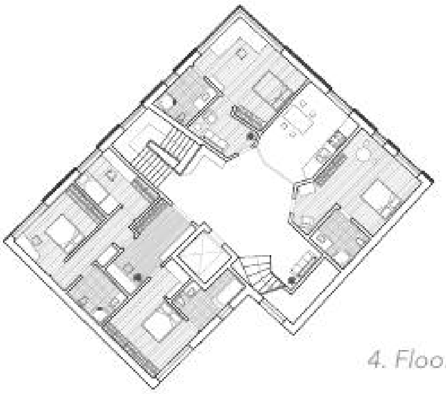
Ground floor



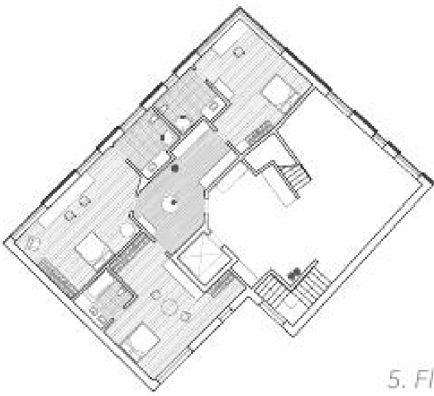
1. Floor

2. Floor

3. Floor

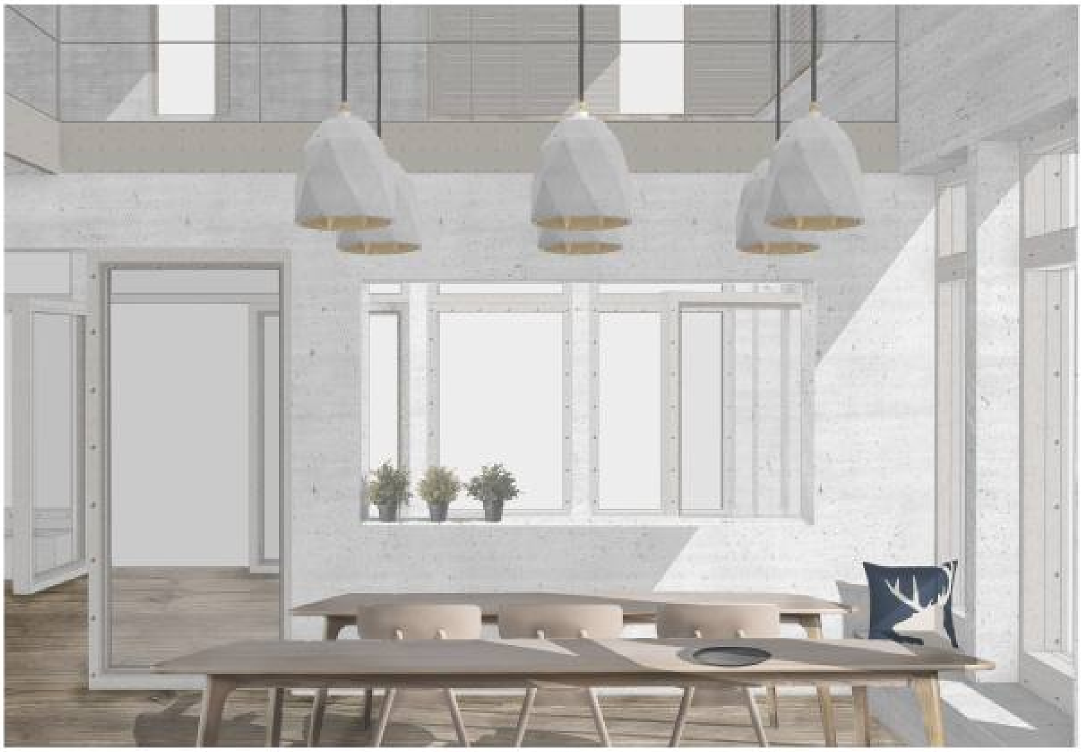


4. Floor



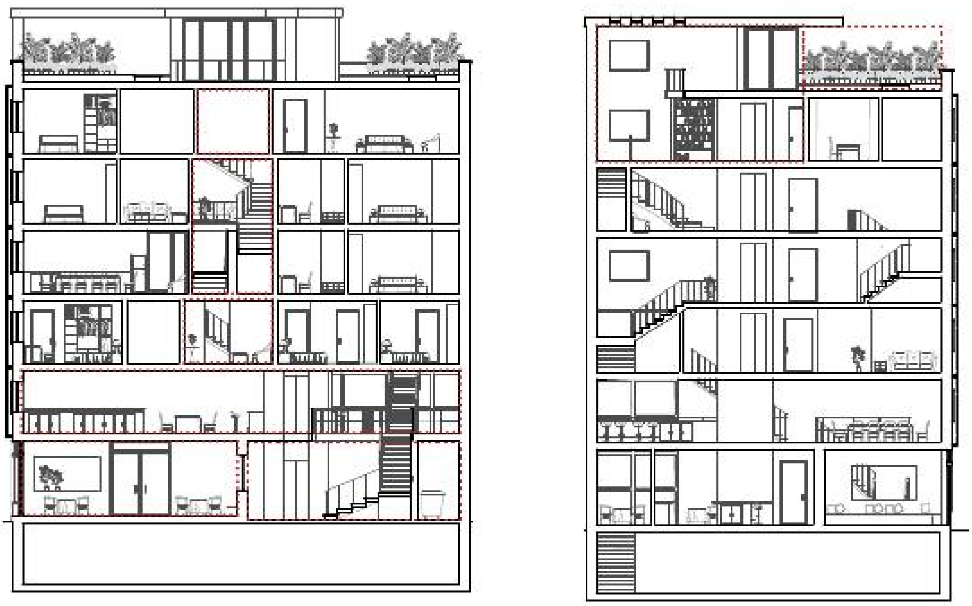
5. Floor

Floor Plans, Original scale 1:100



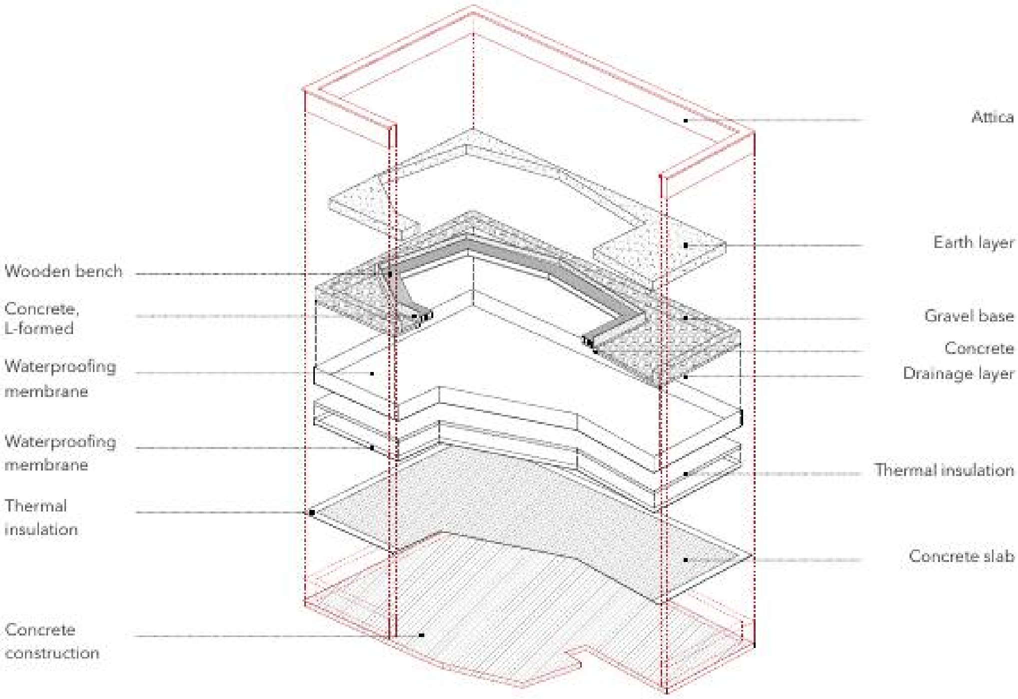
Milieu: the restaurant on the ground floor

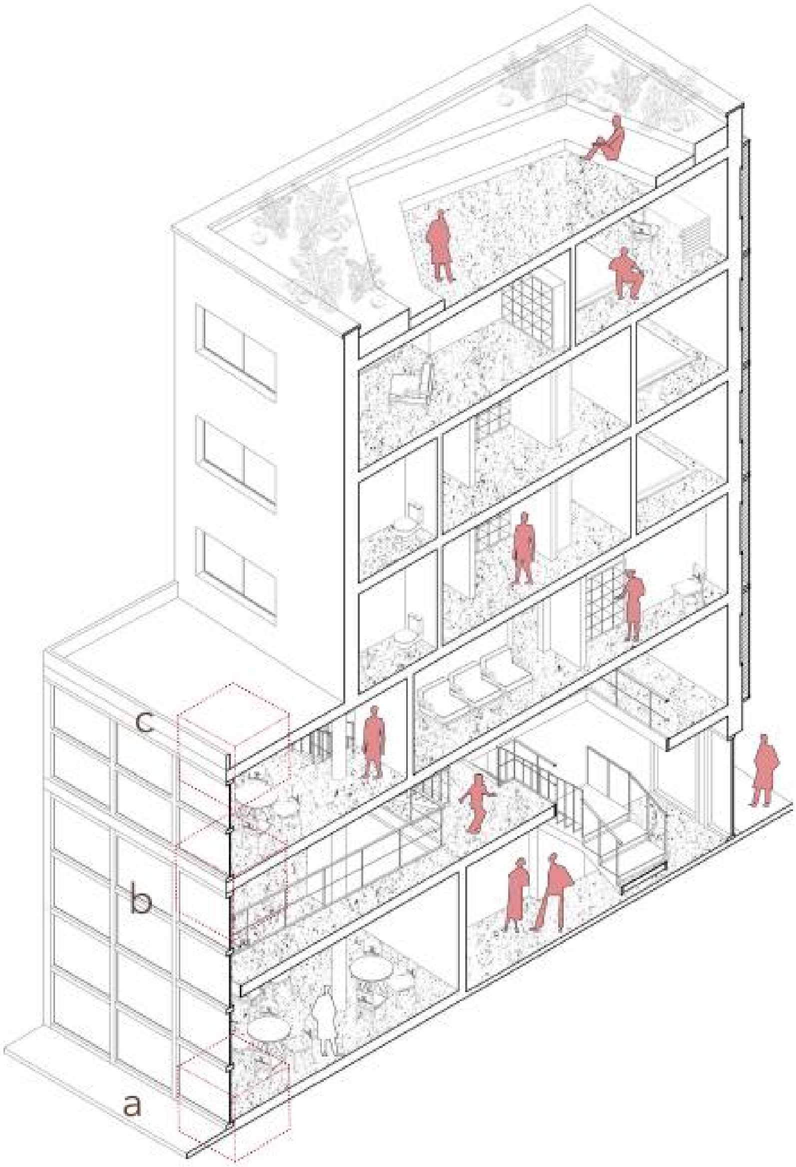
Common spaces shown in the sections

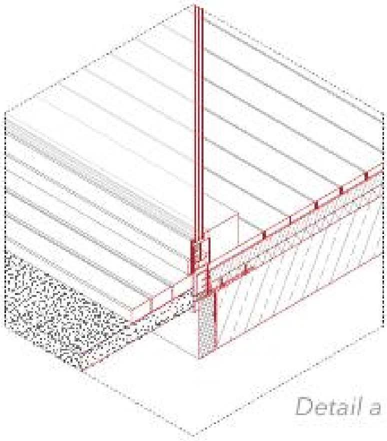


Milieu: the roof

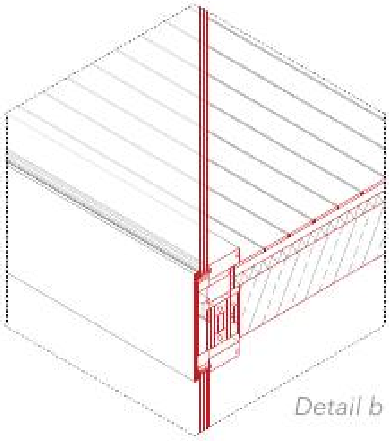
Green roof construction



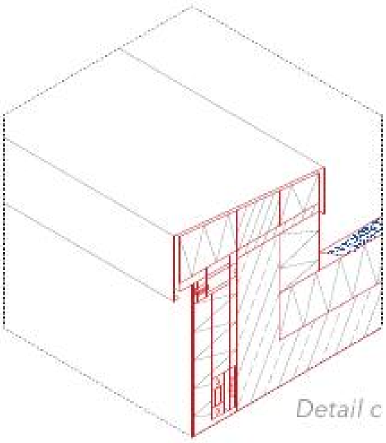




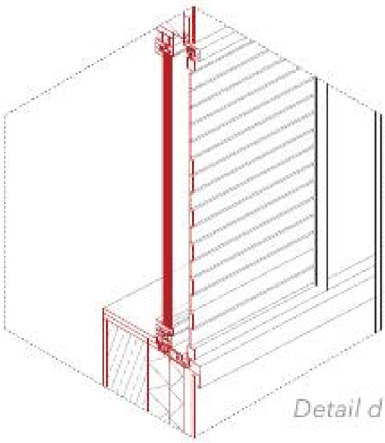
Detail a



Detail b



Detail c



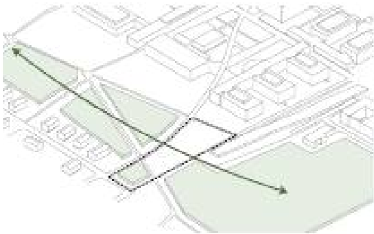
Detail d

02

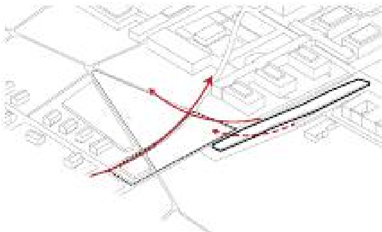
CAMPUS

CAMPUS MARTINSRIED

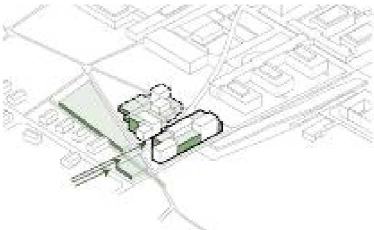
Student Dormitory on LMU Biology Campus
2019 | Chair of Spatial Arts and Lighting Design
Group Work of Two, TU Munich
Martinsried, Germany



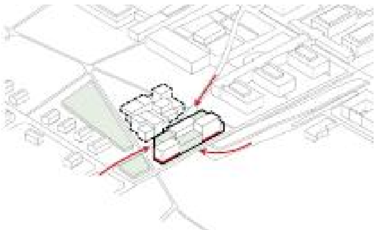
Intersection of nature and architecture



The future subway station and its future connection to the campus



Green as visual and acoustic transmission: Reducing incommings from Martinsried

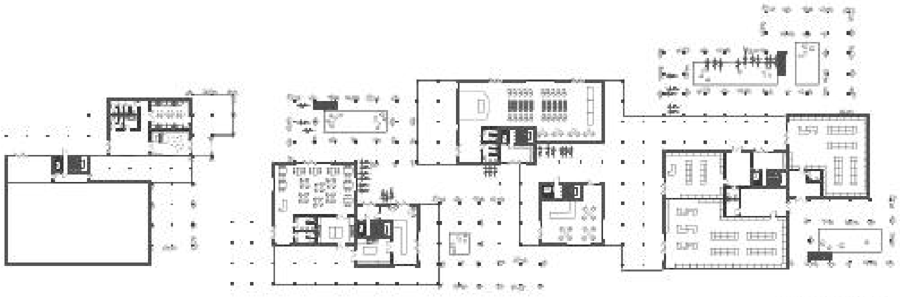


Ground floor as common public space with additional facilities

Site analysis & inspiration

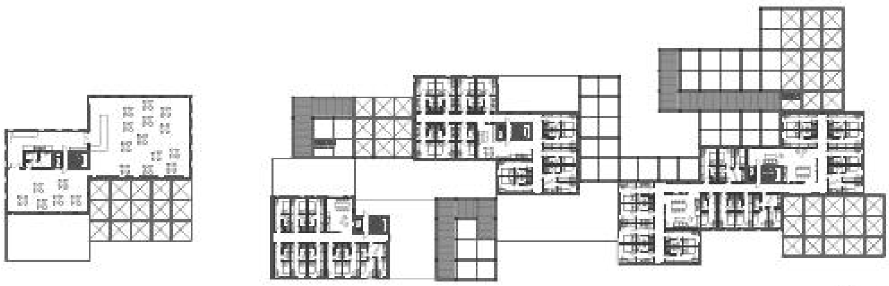
As one of the largest scientific research centers in Europe, Campus Martinsried houses important research facilities, such as the Biomedical Center of the LMU and the Max Planck Institute of Biochemistry. According to the increase in student numbers, a student dormitory needs to be built directly on the campus in order to fulfill the accomodation demand.

By analysing the site, we discovered that the campus, embraced by huge green surface, is at the linking point of district Martinsried and the Biomedical Center of the LMU. This inspires us to integrate green factor in our design to reflect the surrounding within our building and to interprete the relation between the two areas.



Ground floor

1. floor

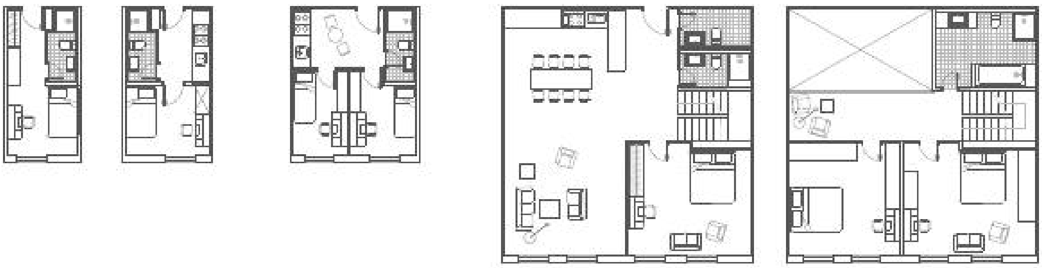


3. floor

Selected floor plans, original scale 1:100

We would like to create a living complex to allow student to have communication with each oterh and also privacy at the same time. Thus, the building volumes should be seperated yet linked to each other. Links should include green volumes, corridors and terracecs. The ground floor is divided into four building blocks for different functions: (from east to west side) block for supermarket and drug store, block for the forum, block for a restaurant and block for sport.

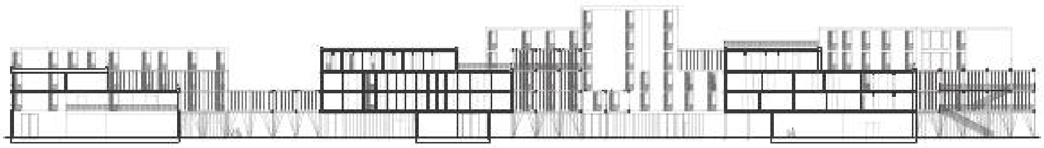
Facade with clinker, original scale 1:20



Milieu: the library on the second floor

Apartment types

In total, there are four appartment types to accommodate students, professors and scientific researchers. Single appartments with no independent kitchens are situated around common kitchens and rooms of each floor. The common rooms are not fully seperated from the rest by walls, but with wood slats and other structures, that allows light to come in. This assures that the corridors are not fully closed, hence, spatial qualities are ensured.



Milieu: the green volumes / spaces
Section OS, original scale 1:100

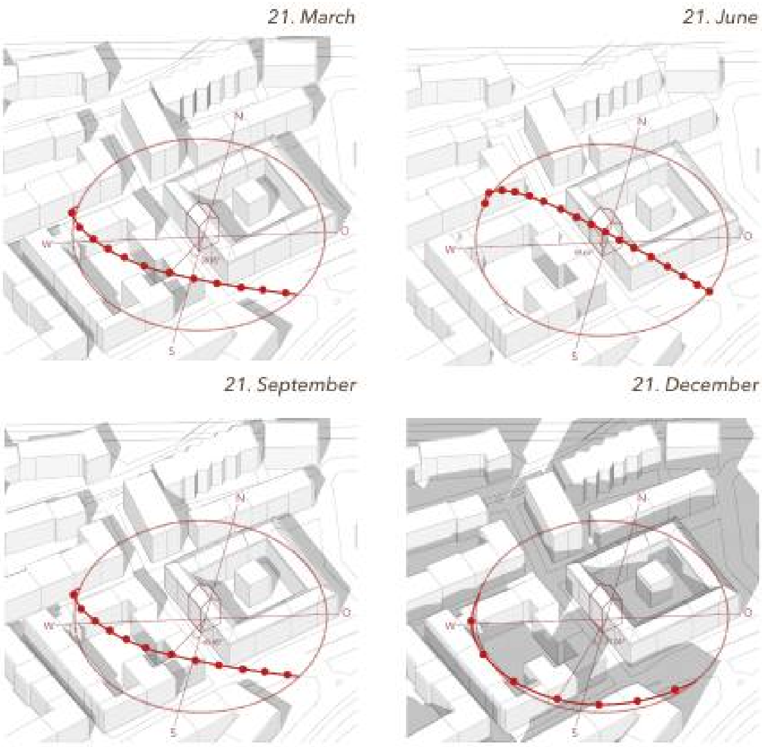
The green volumes is not only considered as connection within the building, the visual and acoustic shelter of the dormitoray, but it also creates a micro climate around the building, and cools down the buidling surface and increase the humidity of the air around the building in the summer.

03

APPARTMENT HOUSE

DIANASTRASSE 4

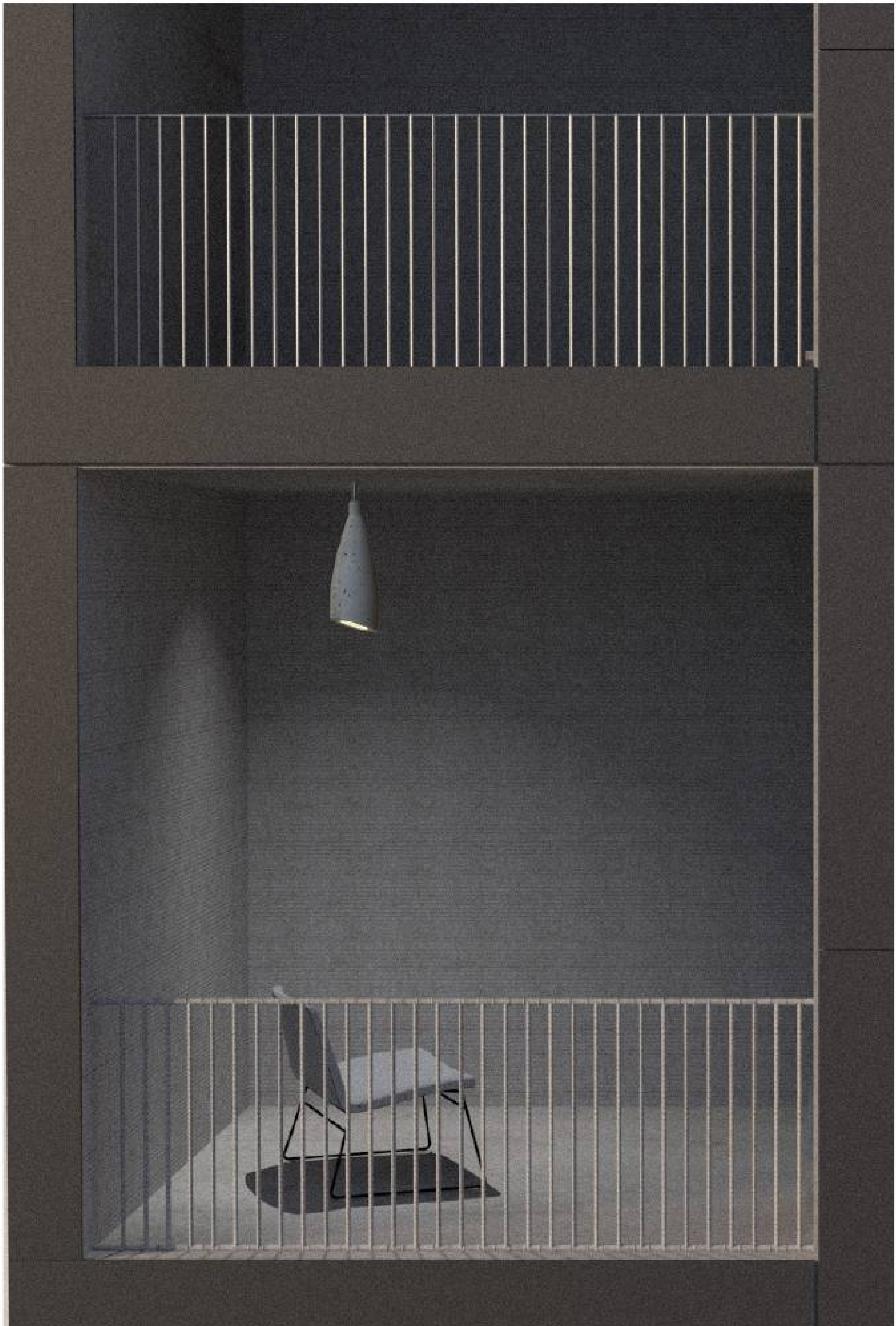
Building Construciton
2015 | Chair of Building Construction & Material Science
Initially Group Work of Three, TU Munich
Munich, Germany



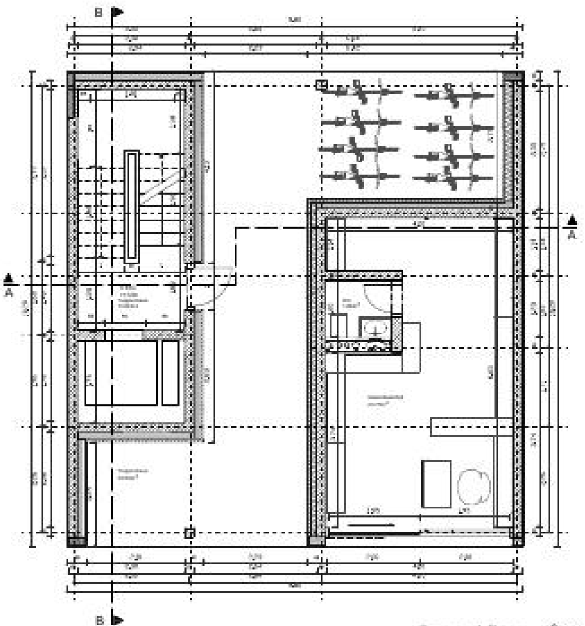
Sun Position Studies

The task of this project is to build an appartment building with a commercial unit in the center of Munich. It should fit in its urban context, and be technologically realisable.

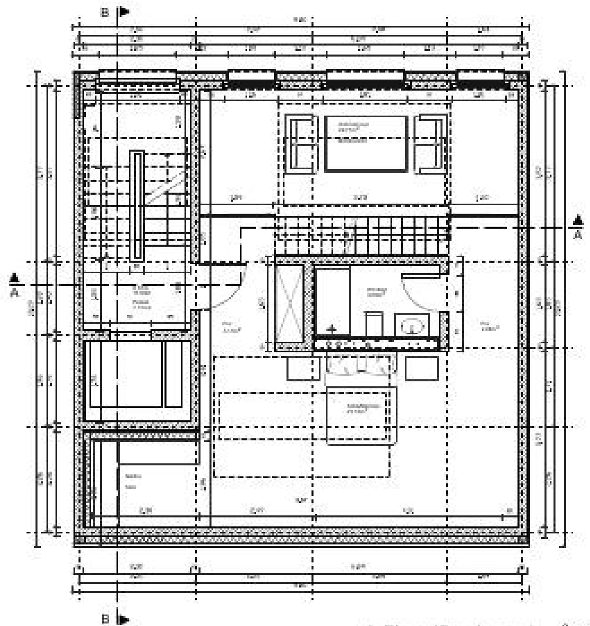
The facade on the street side faces south-west, thus we tried to do the solar analysis at first. According to the sun positions at 12.00 on four particular days of a year, this side of the building gains rich solar energy throughout the year.



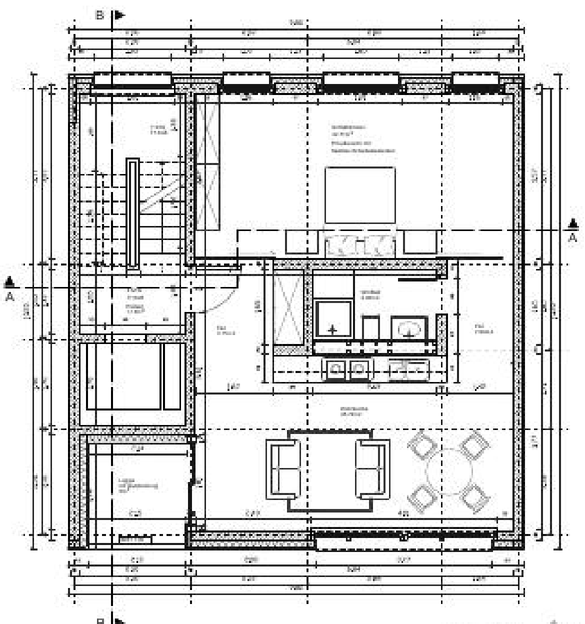
Milieu: living room of the penthouse
Milieu: eat-in kitchen on a regular storey
Floor plans, Original scale 1:50



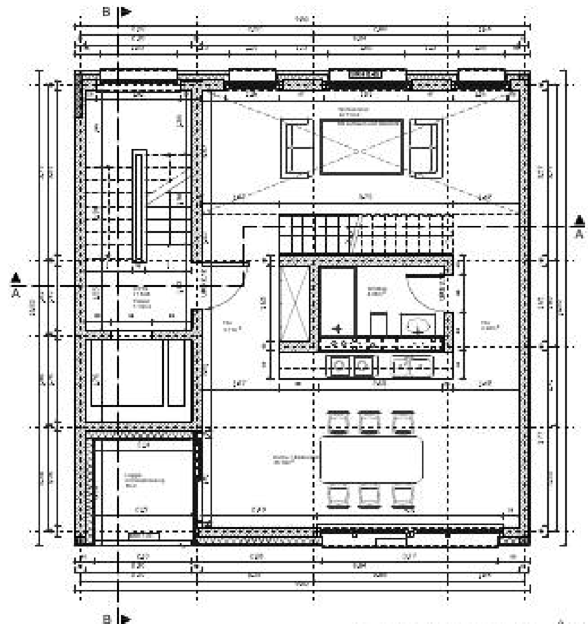
Ground floor



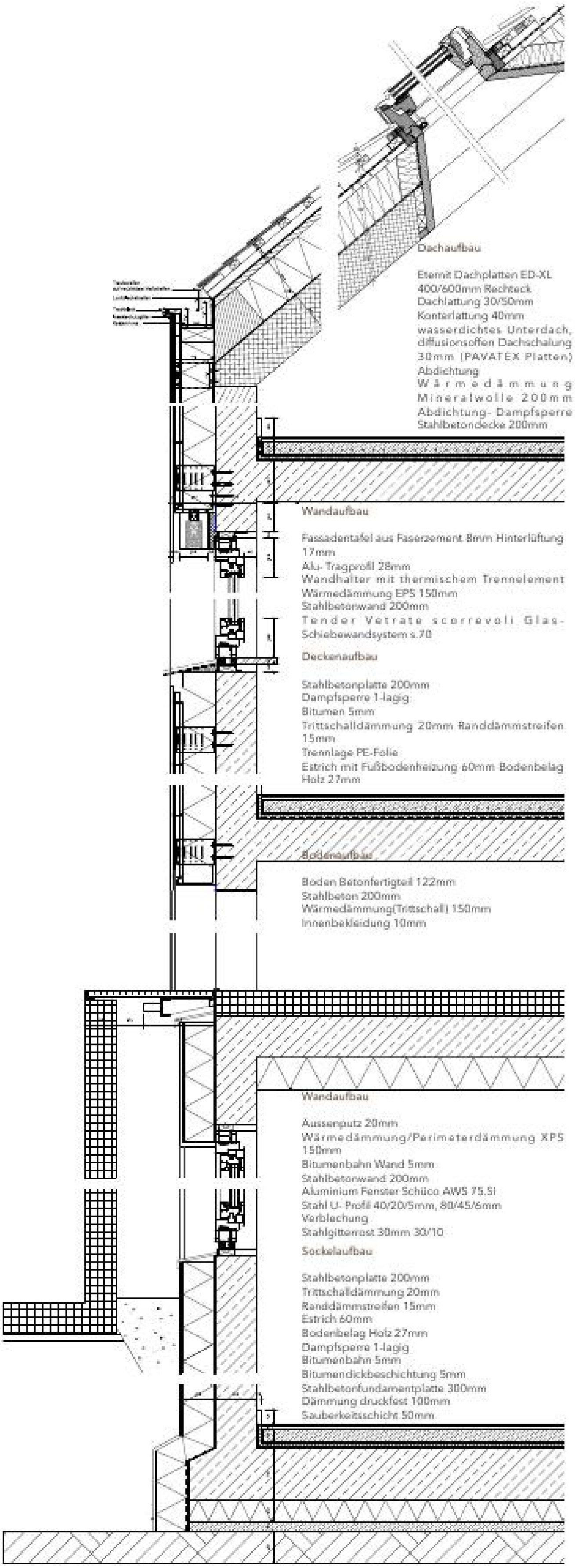
4. Floor (Penthouse)

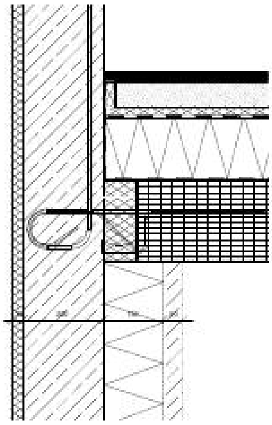


1-3. Floor



5. Floor (Penthouse)



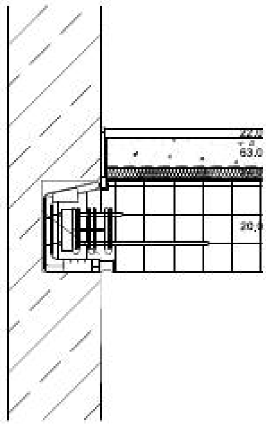


Deckenaufbau

Stahlbetonplatte 200m
1-lagig Bitumen 5mm
Wärmedämmung XPS 1
Trittschalldämmung 20m
PE-Folie
Estrich 60mm
Bodenbelag Holz 27mm

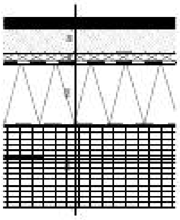
Wandaufbau

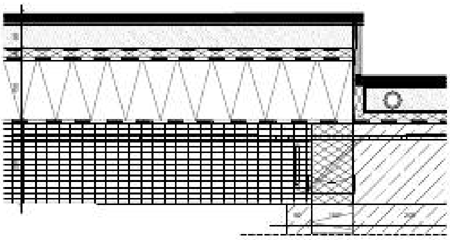
Steinwolle 30mm
Betonwand 200mm
Isokorb Schöck K-WO
Wärmedämmung XPS 1
Betonplatte 50mm

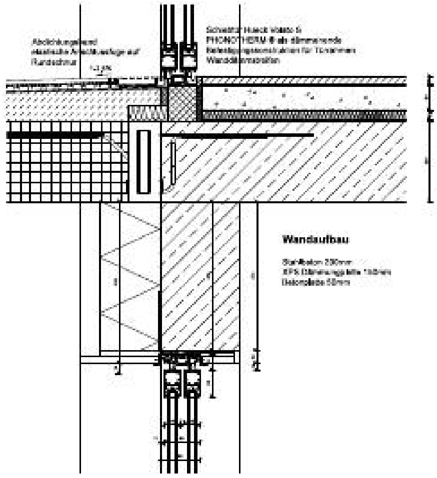


Treppen
mit Schöck-Tronsole zur S

Gelände OKFB 100cm
Holzständer mit Metalbl
in Treppenstufe verschra
Holzhandlauf verzinkt d







Deckenaufbau at the joining point

Isokorb Schöck K-V Stahlbetonplatte 200mm
Dampfsperre 1-lagig
Bitumen 5mm
Wärmedämmung XPS 150mm
Trittschalldämmung 20mm
Randdämmstreifen 15mm
Trennlage PE-Folie
Estrich mit Fußbodenheizung 60mm
Bodenbelag Holz 27mm

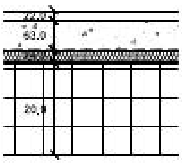
Loggia Tür-/Wandanschluss

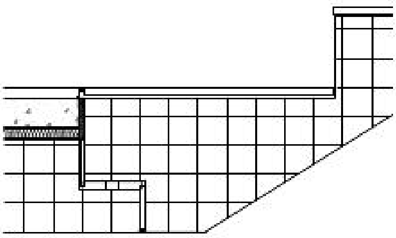
WatecDrainRD-QE Drainrinne
AB-V Abdichtungsband
elastische Abschlussfuge
WatecBW Bewegungsfugenband
Belagsfuge
Betonfertigteil Platte verlegt
Dünnschicht-Drainage 9mm
DiProtec SDB Schnelldichtbahn
Gefälleverbundestrich
Schöck Isokorb
Betonfertigteil 20mm

)0mm Dampfsperre
n
'S 150mm
20mm Trennlage

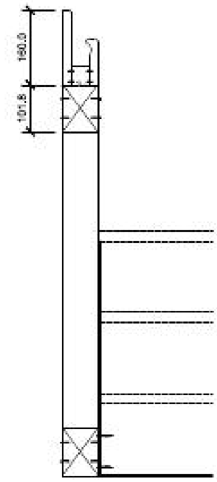
'mm

D
'S 150mm





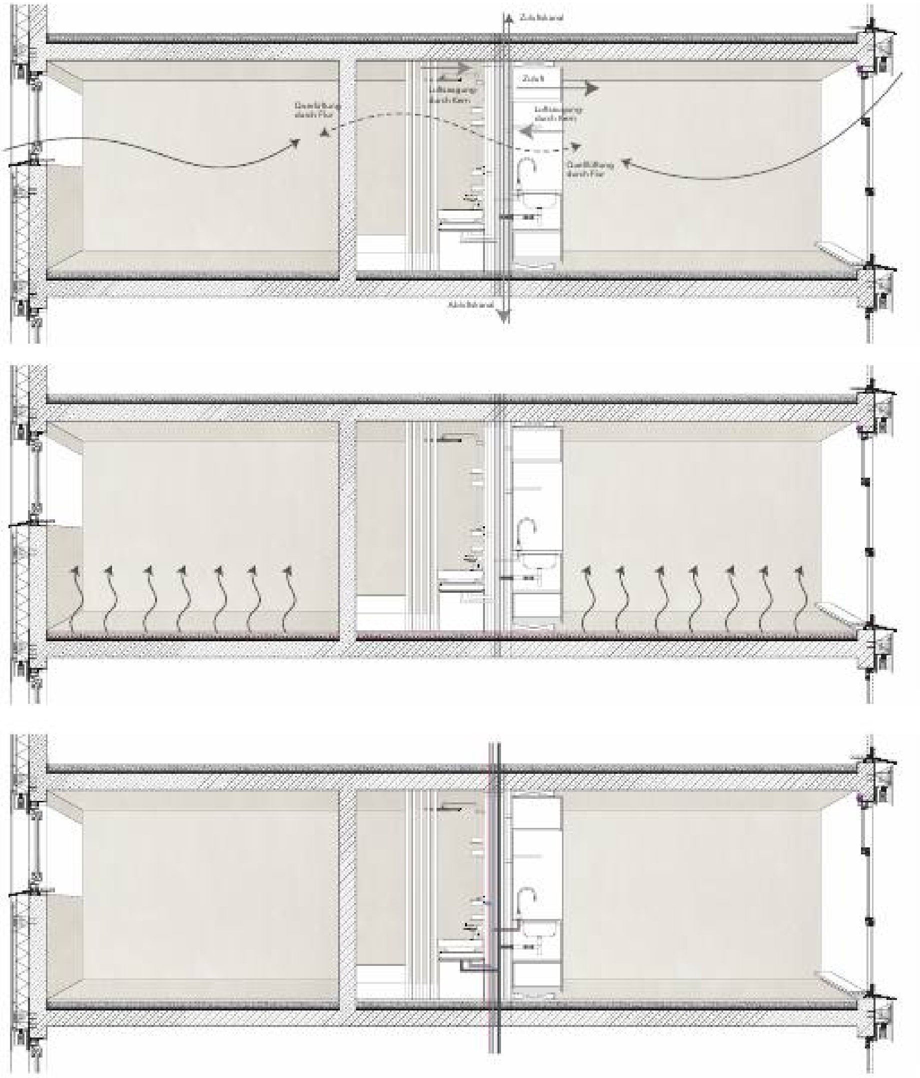
Treppen-wandanschluss M 1:5
mit Schoeck_Tronsole zur Schallentkopplung



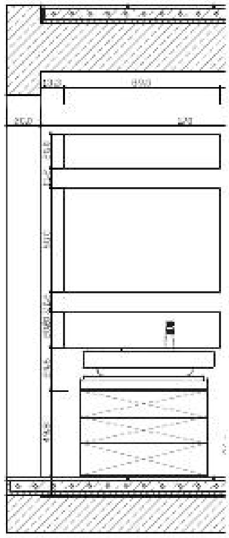
zur Schallentkopplung

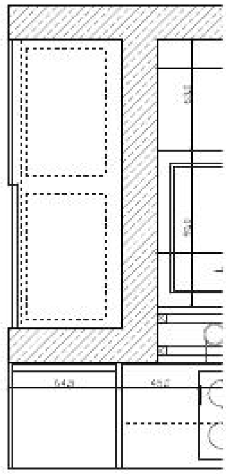
:m
:alblech verkleidet,
:chraubt,
kt darauf gesetzt

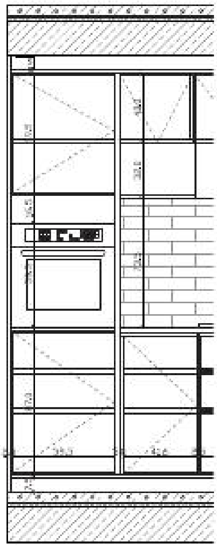
Construction of loggia, Original scale 1:5
Construction of stairs, Original scale 1:5



Ventilation, heating and water systems







HVAC SYSTEM

Since the appartments are not strictly divided by walls, fresh air could be brought inside through natural cross ventilation. In addition to this, there is a centered ventilationsystem, integrated into the installation wall of the sanitary core. Hot water is also transported through pipes in the installation wall and supports the underfloor heating system.

Annual Heating Demand kWh/a

	Area m²	U-Value W/m²K	g-Value
Window South	67,15	0,8	0,63
Window North	43,68	0,8	0,63
Wall	365,15	0,2	-
Cellar ceiling	99,96	0,26	-

QH=QLosses-0,95*QGains=(QT+QL)-0,95*(QS+QI)

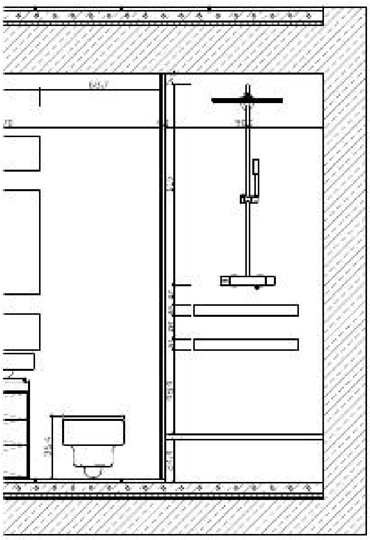
QT=15502,2kWh/a

QL=13522,72kWh/a

QS=2009,17kWh/a

QI=6426,63kWh/a

QH=21011,29kWh/a



Installation wall

Verputzte Wand
Weiss/beige

Spiegel

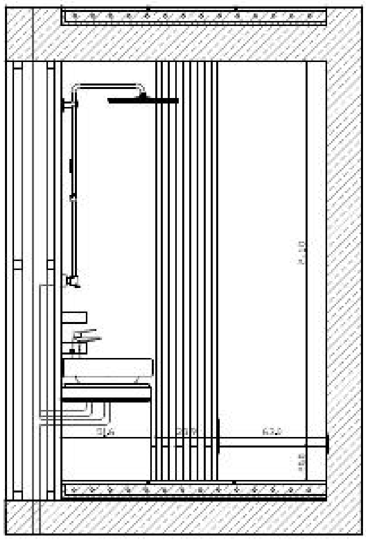
20/89mm, 60/89mm

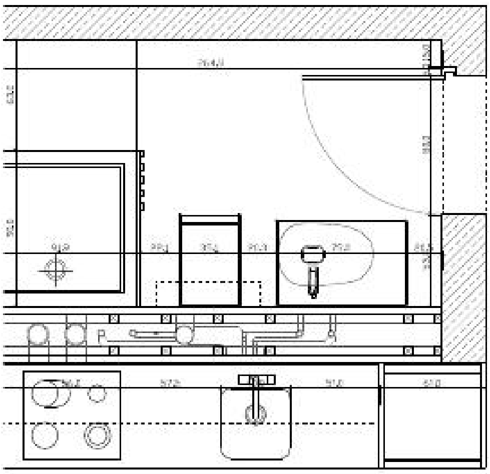
Wasserhahn

Firma GROHE Allure Brilliant
Single-lever basin mixer 1/2"

Waschtisch

Firma Keramag
CITTERIO
Ablagefläche
750/500 mm, ohne Überlauf





Shower

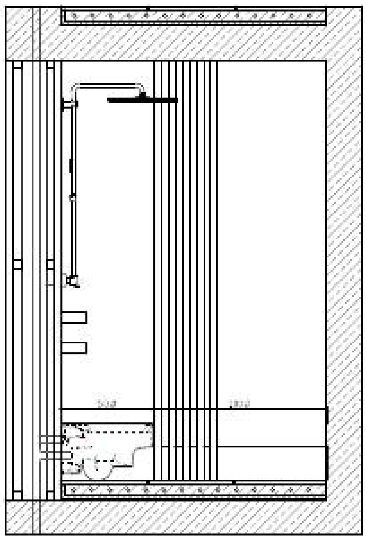
Firma Gohe
Rainshower® System 400
Shower system with thermostat for wall mounting

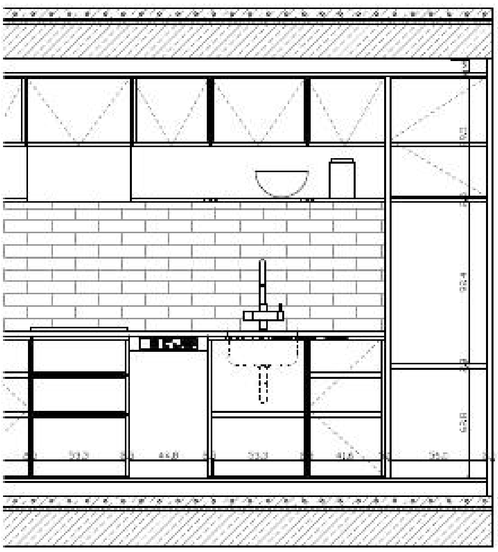
Duschwanne

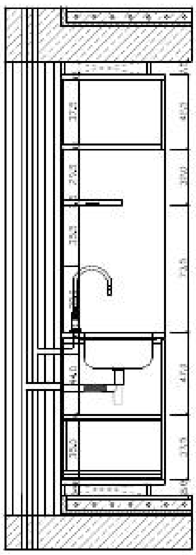
Firma Keramag
Preciosa Duschwanne 900/900 mm, Ablauf: 90mm

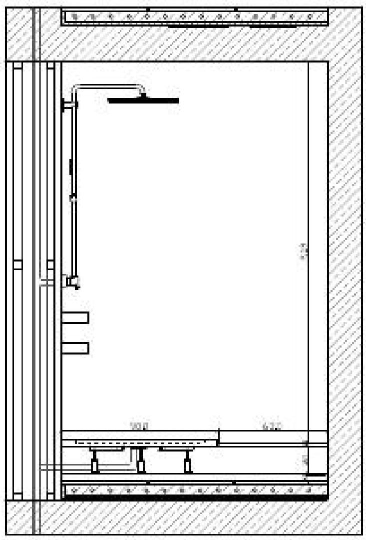
WC

Firma Keramag
Presiosa II Tiefspül-WC, 4,5/6 L, wandhängend 530 mm









04

TOWARDS SOLAR DECATHLON

SWISS LIVING CHALLENGE

Building Performance Studies (Summary)
2017 | Laboratory of Integrated Performance in Design
Group Work of Three, EPFL Lausanne
Fribourg, Switzerland

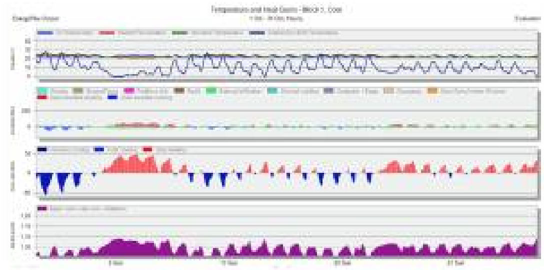
Concept

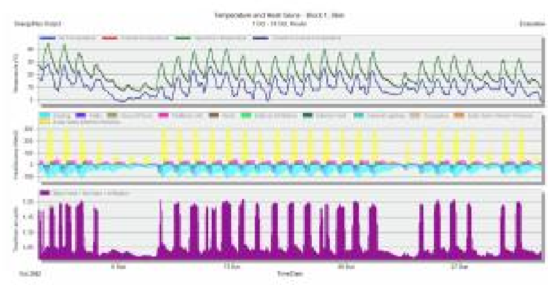
The Solar Decathlon is an international competition. In order to win, students from universities around the globe have to design and build the best life-size scale, fully operational, solar-powered pavilion. This solar housing unit should make at least as much energy as it consumes and should limit its impact on the grid. In order to support the swiss team for the competition in 2017, each group of this course ought to finish a performance study assisted with the simulation software DesignBuilder and provide reasonable proposals.

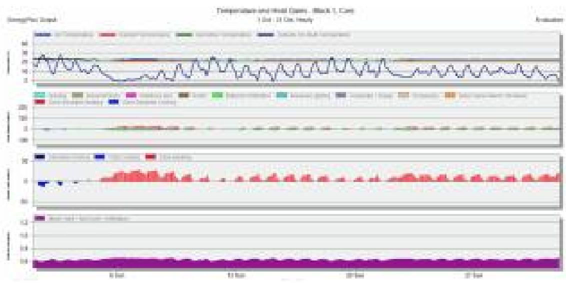
This course finished in January 2017, and the swiss team won the first place in October that year .

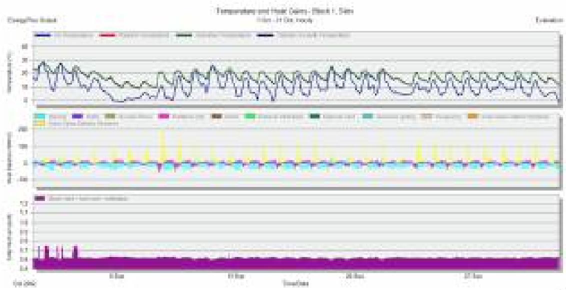
FIRST APPROACH

We proposed to use double glazing with solar protection for both vertical and horizontal windows. This results in lower energy losses due to a lower u-value. The solar protection also results in lower solar gains, which contribute to a lower need for zone cooling, which is also reduced due to the higher construction thickness of 35cm. The LED lighting with controls reduces the total energy consumption to 4.298 kWh/m2, which is a 30% reduction when compared to the base case. Overall, there is lower energy consumption. In terms of lighting we observe that due to incandescent lighting in the base scenario we need less energy for heating, but more electricity is required.









Hourly comfort analysis (October) for the base case, core zone
Hourly comfort analysis (October) for the base case, skin zone

Hourly comfort analysis (October) for our case, core zone
Hourly comfort analysis (October) for our case, skin zone

PASSIVE MEASURES

Following above, we deepened our study. In the passive measures, we suggested to integrate earth as the insulation layer (14cm) of the building envelope, for earth has a relatively high thermal inertia (or volumetric heat capacity). On the roof we planed a green roof and added skylights to reduce the thermal losses through the windows. We then ran simulations with these improvements under deactivated and activated heating conditions to compare the results. Diagramms below show an example comparison under the condition with deactivated heating.

Earth

Termal and Humidity control by pasive behaviour of the material. The earth balances the temperature amortiguating and moving the thermal wave. The same efect ocurs with the humidity. The proposal uses 14cm of earth to reach the 12h delay.

Skylights

Tubular skylights can offer just as much light as zenithal windows in clear conditions and still have a good performance in cloudy conditions, but they have far lower solar gains. This can help us keep the temperature in the core more balanced. Reducing the window size also reduces the losses through it.

Grass

The planted roots mitigate solar radiation by shading the soil and due to the natural cooling effect of evapotranspiration. As an more rapidly growing alternative we choose sedum blanket which, maintaining the outstanding characters of sedum, provides instant ground covering

Thermal comfort with deactivated heating for the base case, core and skin zone

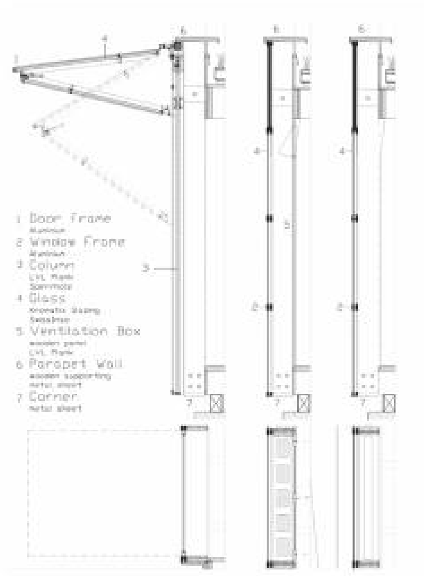
Thermal comfort with integrated inertia layer, skylights and deactivated heatng, core and skin zone

Comfort conditions by scenario for the different building partitions in Denver

Comfort conditions by scenario for the different building partitions in Fribourg

Compared to the highly energy efficient base case we were able to reduce total energy demand by at the most 6-9 kWh/m2 in Denver and Fribourg respectively. Both the People scenario and the Skylights scenario result in large reductions in heating demand, however, for the skylights scenario these are more than offset in the total energy consumption. As explained before this may be due to slightly higher lighting needs when compared to the base scenario, since it was not possible to put in light diffusion parameters in design builder.

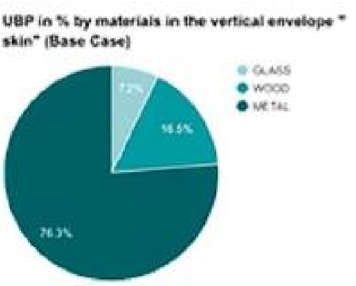
PASSIVE MEASURES - MATERIAL CHOICE

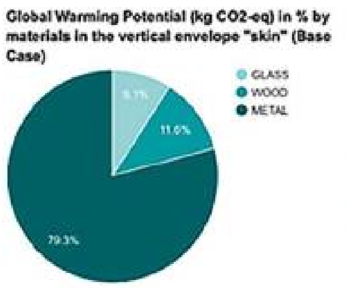


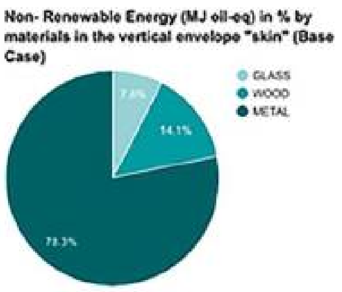
Material	Detail	UBP	Non-Renewable Energy (MJ oil-eq)	Global Warming Potential (kg CO2-eq)
GLASS	Flachglas beschichtet 0.004	2,304,000.00	27,072.00	2,112.00
	Flachglas unbeschichtet 0.004	1,939,200.00	23,616.00	1,893.12
	Flachglas beschichtet 0.012	6,912,000.00	81,216.00	6,336.00
	Flachglas unbeschichtet 0.012	5,817,600.00	70,848.00	5,679.36
	Double glazing	6,873,600.00	83,712.00	6,182.40
	Double glazing with security glass	9,715,200.00	127,296.00	9,158.40
	Triple glazing	12,211,200.00	160,704.00	11,059.20
WOOD	column with LVL Plank	5,292,760.32	50,194.26	2,708.60
	column with Solid Wood	2,696,353.92	24,285.59	1,270.18
	column with OSB	3,669,131.52	44,187.29	2,239.03
METAL	with Galvanized metal sheet	24,415,186.43	279,331.96	18,458.55
	with Titanium zinc	28,724,498.59	279,275.56	18,500.09
	with Aluminium sheet	21,125,949.85	275,087.79	18,286.81

Material	UBP	Non-Renewable Energy (MJ oil-eq)	Global Warming Potential (kg CO2-eq)
Polycarbonat (PC)	2,847,744.00	39,398.40	3,974.40
Flachglas beschichtet 0.004	2,304,000.00	27,072.00	2,112.00
Flachglas unbeschichtet 0.004	1,939,200.00	23,616.00	1,893.12
Flachglas beschichtet 0.012	6,912,000.00	81,216.00	6,336.00
Flachglas unbeschichtet 0.012	5,817,600.00	70,848.00	5,679.36
Double glazing	6,873,600.00	83,712.00	6,182.40
Double glazing with security glass	9,715,200.00	127,296.00	9,158.40
Triple glazing	12,211,200.00	160,704.00	11,059.20

Material	UBP	Non-Renewable Energy (MJ oil-eq)	Global Warming Potential (kg CO2-eq)
Aluminium Frame	20,246,709.00	264,684.69	17,554.95
Wooden Frame	9,011,393.55	86,603.00	5,617.49
PVC Frame	20,597,470.98	224,699.68	14,160.76







Considered building components in our analysis

Calculation results

Going deeper into the material choice, we considered the following elemts in our analysis: the columns consisting of the construction pillars and the aluminium window frame; the ventilation box module; the modulated garage doors (including the aluminium window frame); and the parapet walls (with metallic sheets and wooden supports) and corners. By calculating the values of UBP (environmental damage points, %), Global Warming Potential (kg CO2-eq) and Non-Renewable Energy (MJ oil-eq), we suggest to change the LVL Plank of the wooden constructions into OSB Plank or solid wood and to use polycarbonate glazing instead of normal glass in the window and door areas. Although in reality, materials, such as solid wood, may cause difficulties because of its high cost and weight, which should be considered during the design period.

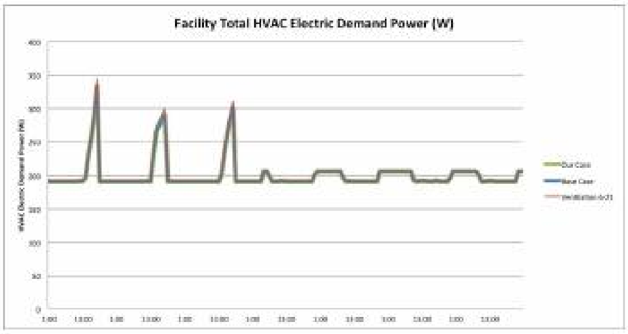
ACTIVE MEASURES

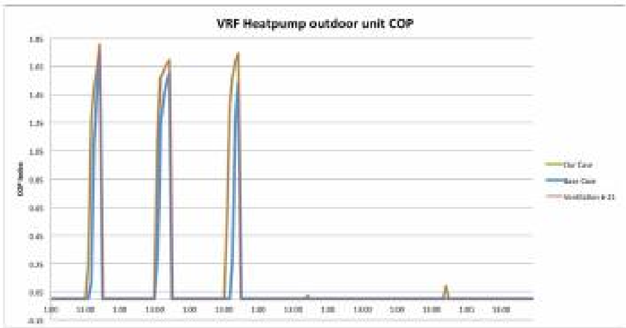
The final part of this report investigates the link between the active system settings and the energetic performance. An analysis is presented of the proposed HVAC System including the solar thermal component for Fribourg, focusing on the effects during spring. In all scenarios the temperature in the core should stay between 18°C and 25°C and the humidity between 35% and 60%, while the light requirement is a minimum of 350 lux. The occupation schedule in the base case is 10 people in the core and 1 person in the skin from 8am to 7pm and 2 people in the core at night from 7pm to 8am.

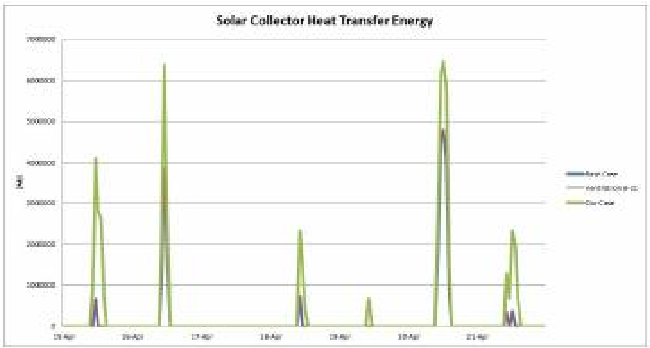
	Total Consumption		Fuel Breakdown						
Variations	*Electricity (Wh/m2)*	*Gas (Wh/m2)*	*Room electricity (Wh/m2)*	*Lighting (kWh)*	*System pumps(Wh/m2)*	*System Fans (Wh/m2)*	*Heating (Wh/m2)*	*Cooling (Wh/m2)*	*Generation Electricity (Wh/m2)*
Base Case	**583.98**		150.83	186.82	209.34	0.11	22.35	**14.53**	-844.34
Ventilation									
Fresh air: 2-Min fresh air (Per person) 8-19 and 19-8 Mon-Sun	**580.55**		150.83	186.82	209.34	0.11	22.35	**11.1**	-844.34
Fresh air: 2-Min fresh air (Per person) 6-21 and off Mon-Sun	**585.85**		150.83	**190.67**	209.34	0.11	**21.79**	**13.12**	-844.34
Fresh air: fix_3-Min fresh air (Per area)	**580.55**		150.83	186.82	209.34	0.11	22.35	**11.1**	-844.34
Temperature Setpoints									
Heating 14C, H set back 12C, Cooling 27C, C set back 29C	**583.98**		150.83	186.82	209.34	0.11	22.35	**14.53**	-844.34
HVAC System									
Heating Gross 6450 COP 3.28	**583.96**		150.83	186.82	209.34	0.11	22.35	**14.51**	-844.34
Cooling Gross 6450 COP 3.28	**583.22**		150.83	186.82	209.34	0.11	22.35	**13.77**	-844.34
Solar Thermal Collector									
THERMOMAX HP450	**583.97**		150.83	186.82	209.34	**0.10**	22.35	**14.53**	-844.34
DHW Tank									
Time settings change: 6-21 Mon-Sun	**583.98**		150.83	186.82	209.34	0.11	22.35	**14.53**	-844.34
Solar Loop									
Time settings change 6-21 Mon-Sun	**610.03**		150.83	186.82	209.34	**0.91**	47.6	**14.53**	-844.34
Increased Tank Size (150%)	**583.97**		150.83	186.82	209.34	**0.10**	22.35	**14.53**	-844.34
Mixed Scenario									
Ventilation: min fresh air per person 24/7; Water Tank Volume: 270L, THERMOMAX; Cooling: Capacity 6450, COP 3.28	**583.2**		150.83	186.82	209.34	**0.09**	22.35	**13.77**	-844.34

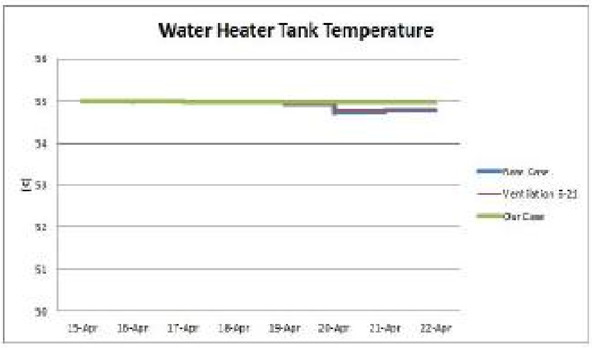
The total and breakdown energy consumption by scenario

We suggest, during the spring period of the 15th to the 21st of April, to use the variable ventilation setting (per person) with the occupancy of 8-19 and 19-8 Mon-Sun (24/7), the original water tank volume of base case (270L), the Thermomax as solar collector which has a better energy storage and transfer capacity, and the cooling system with a larger gross cooling capacity of 6450W and a lower gross rated COP of 3.28. The temperature schedule settings of DHW Tank and Solar Loop System remain the same as in the base case.









Comparison between the base case and our suggestion by scenario

The largest noticeable contribution of the active system was the impact of the solar thermal collector to the water tank system. This is due to the extremely low heating and cooling demand that resulted in the simulation for Fribourg during Spring. The thermal insulation of the solar envelope protected the interior temperature of the core, and significant heat gained came from the occupancy. Our analysis showed that the Thermomax tubular collector, which has a higher overall solar heat transfer efficiency, as well as an in general better performance during cloudy days than the Sebasol collector, would be a better choice.

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